# HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

## PROFESSIONAL EXPERIENCE — 3.5 YEARS

**Software Engineer — HCL Technologies, Chennai, India** Sep 2021 – Jan 2025

- Designed and developed mechanical assemblies and engineered solutions, translating customer specifications into detailed **3D CAD models** and manufacturing documentation.
- Produced engineering calculations and design verification documentation to support product development activities, ensuring adherence to **GD&T;** standards.
- Carried out technical analysis and design reviews for new product development initiatives, contributing to the innovation of engineered solutions.
- Provided technical support to manufacturing and quality teams, troubleshooting production issues and driving **root cause analysis** for process improvements.
- Led engineering project phases from concept through implementation, coordinating with cross-functional teams to achieve project milestones.
- Mentored and supported junior engineers within the team, sharing expertise in mechanical design principles and problem-solving approaches.
- Containerized and deployed legacy Java services onto **AWS EC2** instances using **Docker**, improving deployment consistency and reducing manual setup time by approximately **40%**.

## SKILLS

**Mechanical Design** – Product Development, Engineering Calculations, Design Verification, GD&T;, Technical Documentation, Root Cause Analysis

**CAD Software** – SolidWorks, Inventor

**Engineering Disciplines** – Mechanical Engineering, Manufacturing Engineering, Applied Mathematics, Engineering Mathematics

**Analysis & Simulation** – FEA Analysis, Statistical Analysis, Engineering Analysis

**Cloud & DevOps** – AWS, Docker, CI/CD

**Software Development** – Java, Spring Boot, Python, SQL

## PROJECTS

### Automated Design Verification System

- Developed a Python-based system to automate the generation of design verification reports based on engineering calculation inputs, reducing manual report creation time by **30%**.
- Integrated the system with **Jira** for issue tracking and **Confluence** for documentation, streamlining the change management process for design updates.
- Designed and implemented data parsing logic for various input formats, ensuring compatibility with existing engineering calculation outputs.

### Industrial Control System Component Design

- Modeled and designed custom mechanical components for an industrial control system using **SolidWorks**, focusing on thermal management and vibration resistance.
- Generated detailed engineering drawings and manufacturing specifications, including precise **GD&T;** callouts for a key sensor housing assembly.
- Collaborated with the manufacturing team to refine designs for improved DFM (Design for Manufacturability), resulting in a **15%** reduction in initial assembly defects.

## EDUCATION

**M.Sc. Data Analytics — National College of Ireland, Dublin** Feb 2026

**B.E. Computer Science — SNS College of Technology, Coimbatore, India** 2021

## CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)